

SOFTWARE ENGINEERING LAB - 1 **(UE24CS341A)**

Team Number : 003

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Problem Statement 3 — Campus Placement & Internship Pipeline

Lab 1: Requirements Engineering & UML Use-Case Modelling

Target Stakeholders / Actors: Student Applicant, Placement Officer

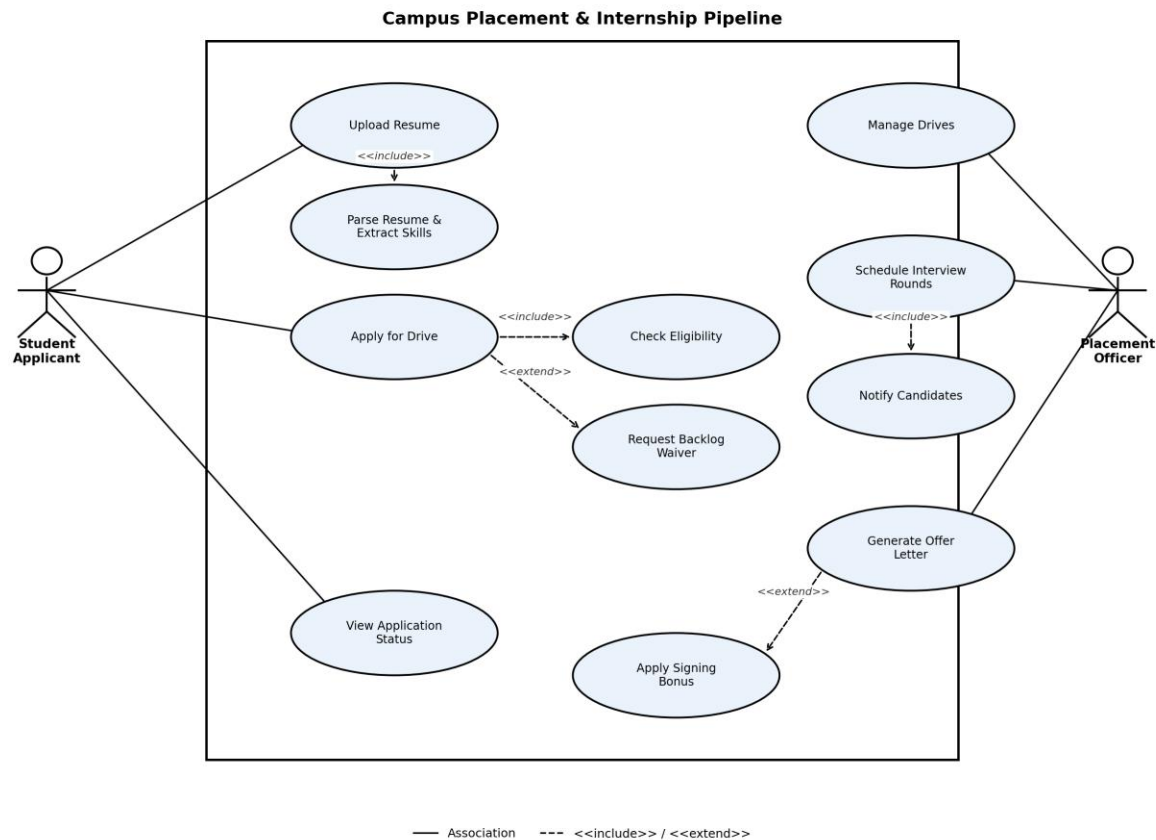
1. Requirements Table

Five Functional Requirements (FR-001–FR-005, with FR-001 as given) and two Nonfunctional Requirements (NFR-001–NFR-002, with NFR-001 as given).

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall parse uploaded student resumes (PDF format), extract skill tags, and match profiles against active company eligibility rules.	High	Pass: Ineligible applicants are filtered out with clear discrepancy reasons. Fail: Applicant with an active backlog bypasses the eligibility filter.	Core resume-screening functionality; given by problem statement.
FR-002	Functional	The system shall allow the Placement Officer to define multi-tier CGPA and backlog filters (e.g., minimum CGPA, max active backlogs) for each company drive.	High	Pass: A drive configured with CGPA ≥ 7.5 and 0 backlogs excludes all applicants below the threshold. Fail: An applicant below the threshold appears in the eligible list.	Drives have varying eligibility criteria that must be configurable per company.
FR-003	Functional	The system shall allow the Placement Officer to create and manage multi-round interview queues, moving eligible applicants sequentially through rounds (e.g., Aptitude, Technical, HR).	Medium	Pass: An applicant who clears Round 1 is automatically queued for Round 2 within 5 minutes. Fail: An applicant remains in the same round queue after being marked as cleared.	Placement drives involve several sequential interview rounds that must be tracked.
FR-004	Functional	The system shall generate and send a digital offer letter to a student once the Placement Officer marks the student as selected for a drive.	High	Pass: Offer letter PDF is generated and emailed to the student within 10 minutes of selection. Fail: Selected student receives no offer letter or an incorrect one.	Offer processing is a core outcome of the placement pipeline.
FR-005	Functional	The system shall display real-time application and interview-round status (Applied, Shortlisted, In-Progress, Selected, Rejected) to the Student Applicant on their dashboard.	Medium	Pass: Status shown on dashboard matches the status last updated by the Placement Officer. Fail: Dashboard shows a stale or incorrect status after an update.	Students need visibility into their application progress without contacting the placement cell.
NFR-001	Nonfunctional	The system shall encrypt all student academic records and offer documents at rest using AES-256.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Protects sensitive student academic and offer data from unauthorized access.
NFR-002	Nonfunctional	The system shall support at least 500 concurrent students submitting applications during peak drive periods with a page response time under 3 seconds.	Medium	Pass: Load testing with 500 concurrent users shows 95th-percentile response time ≤ 3 seconds. Fail: Response time exceeds 3 seconds or requests are dropped.	Placement season creates short bursts of very high concurrent usage that the system must handle without degradation.

2. UML Use-Case Diagram

Actors: Student Applicant and Placement Officer. The diagram below shows all primary use cases along with at least one <<include>> and one <<extend>> relationship (two of each are modelled).



Include relationships: "Upload Resume" includes "Parse Resume & Extract Skills"; "Apply for Drive" includes "Check Eligibility"; "Schedule Interview Rounds" includes "Notify Candidates".

Extend relationships: "Apply for Drive" is extended by "Request Backlog Waiver" (optional, only when eligibility fails on backlog count); "Generate Offer Letter" is extended by "Apply Signing Bonus" (optional, only for select drives).

3. Use-Case Flow Specification

Use Case: Schedule Interview Rounds

Actor: Placement Officer

Precondition: Drive has shortlisted applicants after eligibility check.

Postcondition: Interview round created and candidates notified.

Main Success Scenario:

1. Placement Officer selects a drive's shortlisted applicant list.
2. System displays round options (Aptitude/Technical/HR).
3. Officer defines round date, time, and venue/link.
4. Officer confirms schedule.
5. System creates the round record.
6. System notifies candidates (<<include>>) via email/SMS.
7. Use case ends successfully.

Alternate Flow — 6a. Notification Failure:

6a 1. System detects delivery failure for some candidates.

6a 2. System logs failed recipients and alerts Officer.

6a 3. Officer manually resends or updates contact info; notification retried.